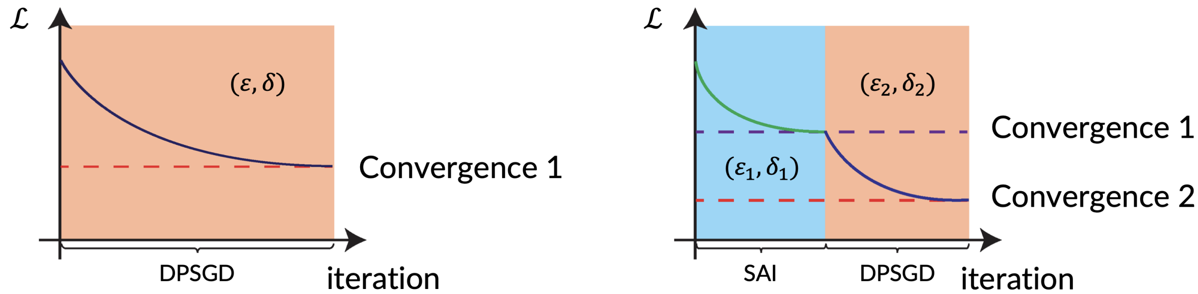

Figure 1: Overview of SAI-DPSGD. SAI-DPSGD utilizes a two-phase approach: the first phase focuses on finding a flatter loss landscape, while the second phase involves training with standard DPSGD for further convergence. The privacy budget (ε, δ) is allocated into $(\varepsilon_1, \delta_1)$ for the sharpness-aware initialization (SAI) and $(\varepsilon_2, \delta_2)$ for DPSGD, with $\varepsilon_1 \gg \varepsilon_2$.

and loss-landscape flatness without auxiliary data remains challenging. Throughout this paper, we use "flatness" to also imply parameter robustness, since a flatter landscape indicates stronger resistance to random perturbations.

Our Solution. A key insight from our work is that flatness typically converges *much earlier* than the training loss. Consequently, we can stop flatness optimization early and then focus on the training loss, avoiding flatness divergence while still achieving sufficient training-loss convergence. Building on this observation, we propose a two-phase approach: *Sharpness-Aware Initialization* (SAI) followed by standard DPSGD (see Figure 1).

By dividing the training process into two distinct phases, we can allocate separate privacy budgets that align with each phase's sensitivity to noise. Since flatter loss landscapes better tolerate noise [56], the DPSGD phase can handle higher noise once flatness has been established. Conversely, optimizing flatness is more sensitive to noise (Section 3.1), justifying a higher budget for the SAI phase. Although DPSGD subsequently runs with relatively higher noise, the landscape remains flat enough to ensure stable convergence. Our theoretical analysis further explains why flatness converges faster than the training loss and how this two-phase strategy improves the privacy-accuracy trade-offs of DPSGD.

The key contribution of SAI-DPSGD lies in its theoretically grounded two-phase framework. While prior research has explored pre-training on public data [70], none has examined a data-driven initialization phase using the same private dataset. Although initialization is crucial in deep neural networks [50], it remains largely unexplored in DPML. We are the first to show that DPML can benefit substantially from a dedicated data-driven initialization phase, which may inspire advanced initialization technologies for further improving DPML. Crucially, our design is not merely empirical; it is guided by rigorous theoretical insights (Section 4.3), providing a robust foundation for future work on even more effective initialization strategies.

Empirical Results. We implemented SAI and evaluated its performance on the MNIST, FashionMNIST, CIFAR-10, CIFAR-100, and SVHN datasets, using a variety of model architectures. These include CNN-Tanh and DP-NASNet [15], both designed specifically for DPML, GNResNet with group normalization [73] (a common replacement for batch normalization in DPML [8,10,19,67,71]), and vision transformers (ViTs) [21] pretrained on ImageNet. Our results consistently show stable accuracy improvements compared to state-of-the-art approaches. For example, SAI-DPSGD improves the accuracy of CNN-Tanh with SELU on CIFAR-10 by over 6% for $\varepsilon = 1$, whereas most existing DPML enhancements achieve gains of less than 3% [71] under the same privacy budget.

Contributions. Our key contributions are summarized below:

• *New technique*. We propose SAI, a method that improves DPML accuracy by promoting a flatter loss landscape. The SAI-DPSGD framework achieves better privacy-accuracy trade-offs compared to existing approaches.

• *Theoretical understandings*. We analyze SAI theoretically to show why it is effective, revealing a fundamental trade-off between minimal flatness and training-loss convergence. This analysis also identifies a theoretical optimum range that guides hyperparameter choices.

• *Extensive empirical studies*. We conduct comprehensive experiments on SAI-DPSGD across diverse tasks and privacy budgets. Results show that SAI-DPSGD consistently improves the privacy-accuracy trade-off on standard DPML benchmarks and across various scenarios.

2 Background

2.1 Differentially Private Machine Learning

Differential privacy [24] (DP) is a mathematical framework designed to protect individuals' privacy when analyzing and sharing sensitive data. It allows for the extraction of useful statistical information from a population while minimizing the risk of exposing any individual's private information. The core concept of DP is to introduce a controlled amount of noise